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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 07/03/2007 |
| **PI Name:** | Derby, Jeffrey J. Ph.D. | | |
| **Project Title:** | Quantitative Models for Microgravity Melt Crystal Growth with Additional Applications to Detached Solidification and II-VI Materials | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Basics of solidification | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 55455-0132 | **Congressional District:** | 5 |
| **Comments:** | | | |
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| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
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| **Key Personnel Changes/Previous PI:** | | | |
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Large single-crystals of exacting quality are needed in the fabrication of advanced optical, magnetic, and electronic devices, and the growth of these crystals presents some of the most difficult challenges today in materials processing. Crystal growth conducted under microgravity conditions has had a profound impact on improving our understanding of these systems, and the research proposed here aims to further this knowledge. The ground-based research program undertaken here will develop powerful, new computational models for the analysis and optimization of microgravity crystal growth processes. These models will couple a realistic and user-friendly model for furnace heat transfer with a powerful finite element code for computing continuum transport and interfacial phenomena in crystal growth. Parallel with this work, these models will be applied to study phenomena of great significance in melt crystal growth, specifically three-dimensional, transient effects on melt convection and segregation during the growth of II-VI materials. A material of particular interest is cadmium zinc telluride (CZT), an important strategic material used in portable, high-performance

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| | radiation detectors. |
| **Task Description:** | Our immediate objectives involve model and algorithm development to couple two independent heat transfer codes representing different scales and phenomena in a melt crystal growth system (a "global" furnace-level model and a "local" model that represents phenomena relevant to solidification) via shared domain boundaries. In general, such strategies are attractive because of their simplicity for solving problems that may be represented by linking existing codes; however, they often meet with only limited success due to convergence difficulties associated with the overall model. Our challenge is to carry out this integration using an algorithm that is self-consistent, rigorous, efficient, and robust.<br><br>The application of crystal growth models to quantitatively analyze phenomena or for process design and operation has been relatively limited. Although the most sophisticated of these models realistically capture the non-linear interaction between bulk transport, phase change, and thermodynamic partitioning of components between melt and the crystal, to date few models have included a realistic account of furnace heat transfer. Instead, furnace heat transfer is usually idealized in terms of a simple furnace temperature profile or heat fluxes imposed as boundary conditions. In many cases, such an idealization limits the model to predicting only qualitative trends. Ultimately such models are unable to represent important interactions between furnace heat transfer and local continuum transport phenomena. An understanding of such interactions is critical to the improvement of these crystal growth techniques, because furnace design is one of the few means by which the crystal grower can influence the growth process. This work is developing a multi-scale model to simulate realistic crystal growth systems by coupling two different modeling tools using an iterative scheme. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | In addition to its microgravity relevance, this work promises many other significant benefits to ground-based research and production of crystals grown from the melt. In general terms, the understanding gained from successful modeling of crystal growth systems will lead to better process operation and design, ultimately yielding better quality crystals at higher production rates and lower costs. In addition, the computational challenges of realistically modeling these systems are significant, and the continued development of algorithms for high performance computing will be noteworthy in itself. This work will also feature significant international collaboration, outreach, and academic achievement. This research will be conducted with international partners, promising a symbiosis of cultural and research approaches that will benefit both partners and nations in the areas of microgravity and materials science research and education. Finally, the multi-disciplinary and technologically relevant nature of this work promises to impart the graduate students involved with modern skills needed to advance materials processing research and practice of tomorrow. |
| **Task Progress:** | We have made significant progress on two fronts. We have developed a new, higher-order finite element solution approach for incompressible fluid flow. This method is based on an unstructured, tetrahedral mesh and is implemented using a Galerkin approach with mixed representation for velocity and pressure fields. Preliminary results indicate that this approach will yield an order of magnitude increase in solution accuracy for comparable effort compared to equal-order, pressure-stabilized finite element implementations. We have also developed a successful approach to link a furnace model to a local crystal growth model in a self-consistent formulation. Such an approach is necessary for quantitative modeling of melt crystal growth systems and is ideally suited for the analysis of microgravity crystal growth experiments. |
| **Bibliography Type:** | Description: (Last Updated: 09/04/2025) |
| **Articles in Peer-reviewed Journals** | Sonda P, Yeckel A, Daoutidis P, Derby JJ. "Hopf bifurcation and solution multiplicity in a model for destabilized Bridgman crystal growth." Chemical Engineering Science, 2005 Mar;60(5):1323-60. http://dx.doi.org/10.1016/j.ces.2004.10.006 , Mar-2005 |
| **Articles in Peer-reviewed Journals** | Pandy A, Yeckel A, Reed M, Szeles C, Hainke M, Müller G, Derby JJ. "Analysis of the growth of cadmium zinc telluride in an electrodynamic gradient freeze furnace via a self-consistent, multi-scale numerical model." Journal of Crystal Growth 2005 Mar 15;276(1-2):133-47. http://dx.doi.org/10.1016/j.jcrysgro.2004.11.303 , Mar-2005 |
| **Articles in Peer-reviewed Journals** | Lun L, Yeckel A, Reed M, Szeles C, Daoutidis P, Derby JJ. "On the effects of furnace gradients on interface shape during the growth of cadmium zinc telluride in EDG furnaces." Journal of Crystal Growth 2006 Apr 15;290(1):35-43. http://dx.doi.org/10.1016/j.jcrysgro.2006.01.006 , Jun-2006 |
| **Articles in Peer-reviewed Journals** | Yeckel A, Pandy A, Derby JJ. "Fixed-point convergence of modular, steady-state heat transfer models coupling multiple scales and phenomena for melt-crystal growth." Int J Numer Methods Eng 2006 Sep 17; 67(12):1768-89. http://dx.doi.org/10.1002/nme.1685 , Sep-2006 |
| **Articles in Peer-reviewed Journals** | Lun L, Yeckel A, Daoutidis P, Derby JJ. "Decreasing lateral segregation in cadmium zinc telluride via ampoule tilting during vertical Bridgman growth." Journal of Crystal Growth 2006 Jun 1;291(2):348-57. http://dx.doi.org/10.1016/j.jcrysgro.2006.03.019 , Jun-2006 |
| **Articles in Peer-reviewed Journals** | Derby JJ, Kwon YI, Pandy A, Sonda P, Yeckel A, Jung T, Muller G. "Developing quantitative, multiscale models for microgravity crystal growth." Ann N Y Acad Sci. 2006 Sep;1077:124-45. PMID: 17124119 , Sep-2006 |
| **Articles in Peer-reviewed Journals** | Derby JJ, Lun L, Yeckel A. "Strategies for the coupling of global and local crystal growth models." Journal of Crystal Growth 2007 May 1;303(1)114-23. http://dx.doi.org/10.1016/j.jcrysgro.2006.11.343 , May-2007 |
| **Books/Book Chapters** | Yeckel A, Derby JJ. "Computer modelling of bulk crystal growth." in "Bulk Crystal Growth of Electronic, Optical, and Optoelectronic Materials." Ed. P. Capper. Chichester ; Hoboken, NJ : John Wiley & Sons, c2005., Jan-2005 |
| **Books/Book Chapters** | Derby JJ. "Crystal Growth: Crystal growth, bulk (Theory and Models)" in "Encyclopedia of Condensed Matter Physics." Ed. F. Bassani, G.L. Wyder, J. Liedl. Amsterdam ; Boston : Elsevier, 2005., Jan-2005 |

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| **Books/Book Chapters** | Derby JJ, Kwon YI, Pandy A, Sonda P, Yeckel A, Jung T, Müller G. "Developing Quantitative, Multi-Scale Models For Microgravity Crystal Growth." in "Interdisciplinary transport phenomena in the space sciences. Proceedings of the Interdisciplinary transport phenomena in the microgravity space sciences IV conference, Tomar, Portugal, August 2005." Ed. S.S. Sadhal et al. Boston, Mass. : Published by Blackwell Pub. on behalf of the New York Academy of Sciences, 2006., Sep-2006 |